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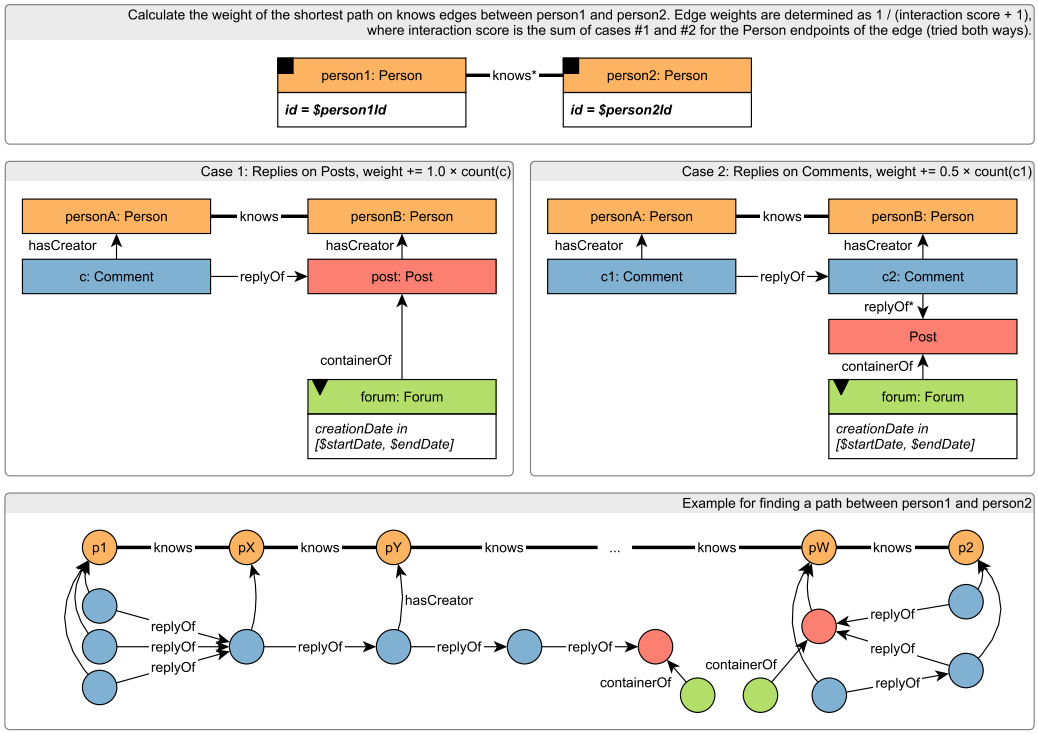
query	BI / read / 15
title	Trusted connection paths through forums created in a given timeframe
pattern	

description	Given two Persons with IDs `$person1Id` and `$person2Id`, calculate the cost of the weighted shortest path between these two Persons, in the subgraph induced by the knows relationship. The interaction score of a knows edge is calculated based on the interactions of its Person endpoints: • Every direct reply (by one of the Persons) to a Post (by the other Person) is 1.0 point. • Every direct reply (by one of the Persons) to a Comment (by the other Person) is 0.5 points. Only consider Messages that were created in a Forum that was created within the timeframe (interval) `[$startDate, $endDate]`. Note that for Comments, the containing Forum is that of the Post that the comment (transitively) replies to. Also note that interactions are counted both ways. The weight for the shortest path algorithm is determined as $\frac{1}{interaction\ score+1}$. The result of the query is a single number, the cost of the weighted shortest path. If no such path exists, the query should return −1.0.

params

1	`$person1Id`	ID	(a) `$person1Id` – `$person2Id` pair with a distance of 4 hops (b) `$person1Id` – `$person2Id` pair with a distance of 2 hops
2	`$person2Id`	ID	
3	`$startDate`	Date	(a) Small interval (approx. one week) (b) Big interval (approx. one month)
4	`$endDate`	Date	

result

1	`weight`	32-bit Float	C	

limit	n/a
CPs	1.2, 2.1, 2.2, 2.4, 3.3, 5.1, 5.3, 7.2, 7.3, 7.6, 7.7, 8.1, 8.2, 8.3, 8.4, 8.5, 8.6